Informatica

Prerequisites to connect to a serverless SQL pool

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Abstract

You can connect to a serverless SQL pool when you read data from Microsoft Azure Synapse SQL. This article explains the prerequisite tasks that you must complete before you configure a Microsoft Azure Synapse SQL connection to connect to a serverless SQL pool.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector

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Overview

Serverless SQL pool allows you to read data from a file in Microsoft Azure Data Lake Storage Gen2 using an OPENROWSET query or by creating an external table of the file.

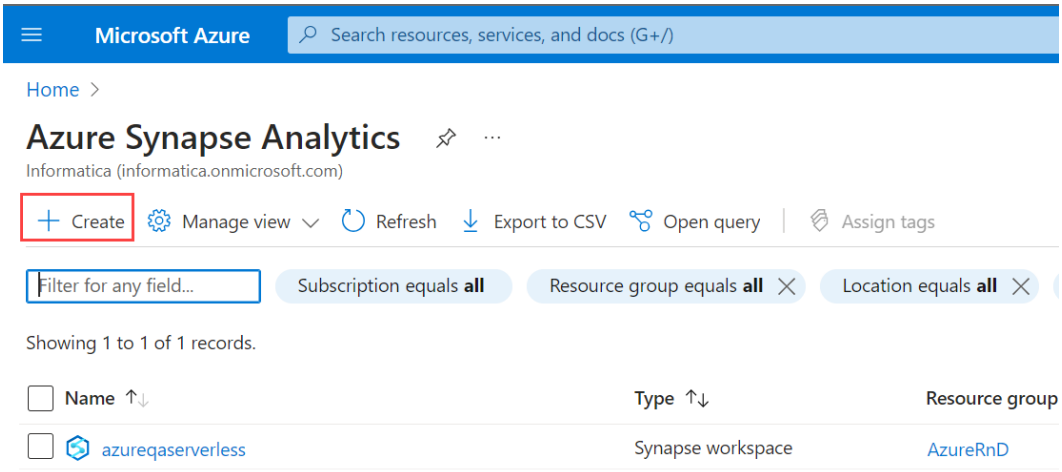
Before you connect to a serverless SQL pool, complete the following prerequisite tasks:

1. Configure an Azure Analytics serverless pool workspace.
2. Create an SQL database in the serverless pool workspace.
3. Configure the following authentication types:
   - Microsoft SQL Server authentication
   - Azure Active Directory (AAD) authentication
   - Managed Identity authentication
   - Service Principal authentication
4. Obtain credentials for Service Principal Authentication to connect to Microsoft Azure Data Lake Storage Gen2 to stage the files.
5. Configure steps to read data from a file using the OPENROWSET query or by creating an external table.
6. Create a Microsoft Azure Synapse SQL connection.
   For more information about configuring a Microsoft Azure Synapse SQL connection, see the *Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector documentation*.

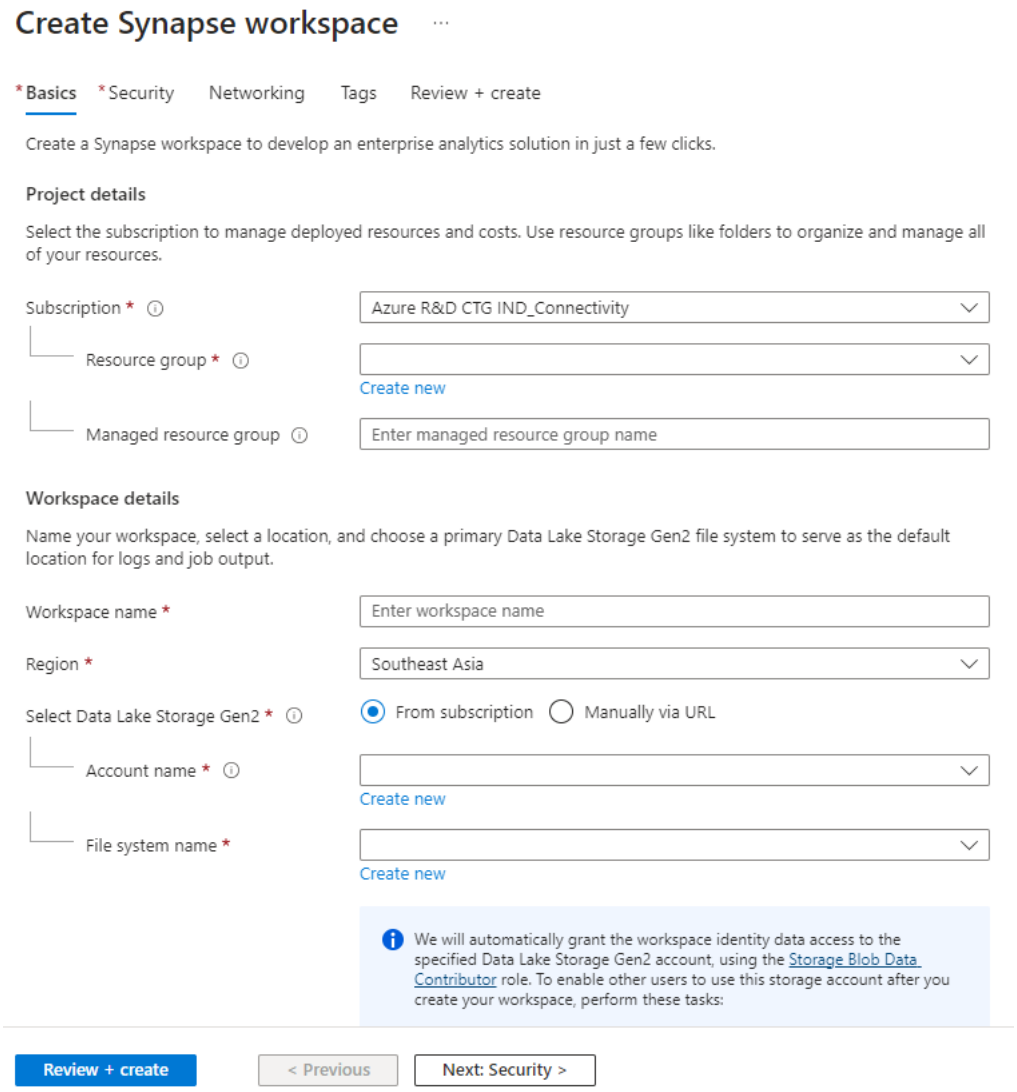
Creating a workspace

Create a Synapse workspace in your Azure account.

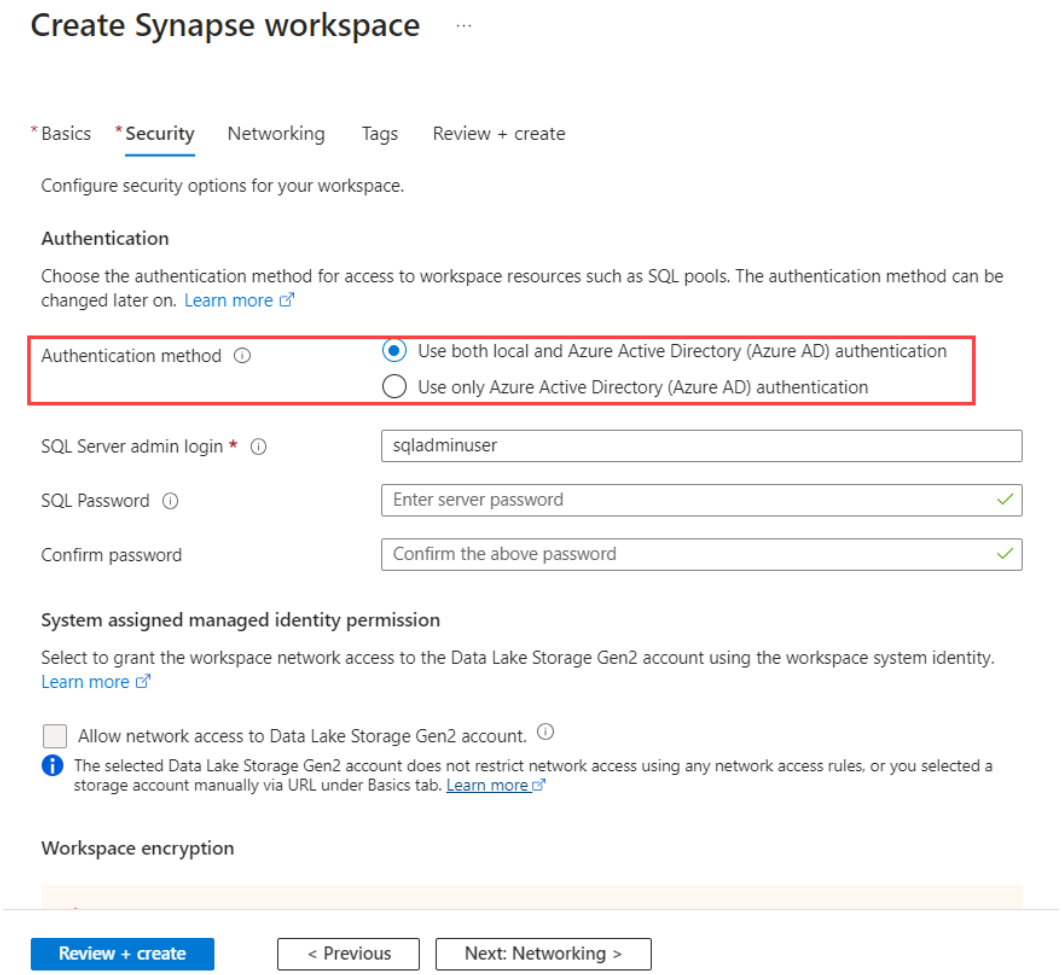
1. Log in to the Azure portal.
2. Search for Azure Synapse Analytics.
3. Click **Create** to create a new Synapse workspace.



4. On the **Create Synapse workspace** page, enter the details for the new workspace.



   a. In the **Subscription** field, select the subscription in which you want to create the workspace.
   b. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.
   c. In the **Workspace name** field, enter a name for the workspace.
   d. In the **Region** field, select a location for the workspace.
   e. Select the **From subscription** option to access the Data Lake Storage Gen2 account from the subscription.
   f. Select the Data Lake Storage Gen2 **Account name** that belongs to the subscription and **File system name** that belongs to the account.
5. Navigate to the **Security** tab and configure security options for your workspace.

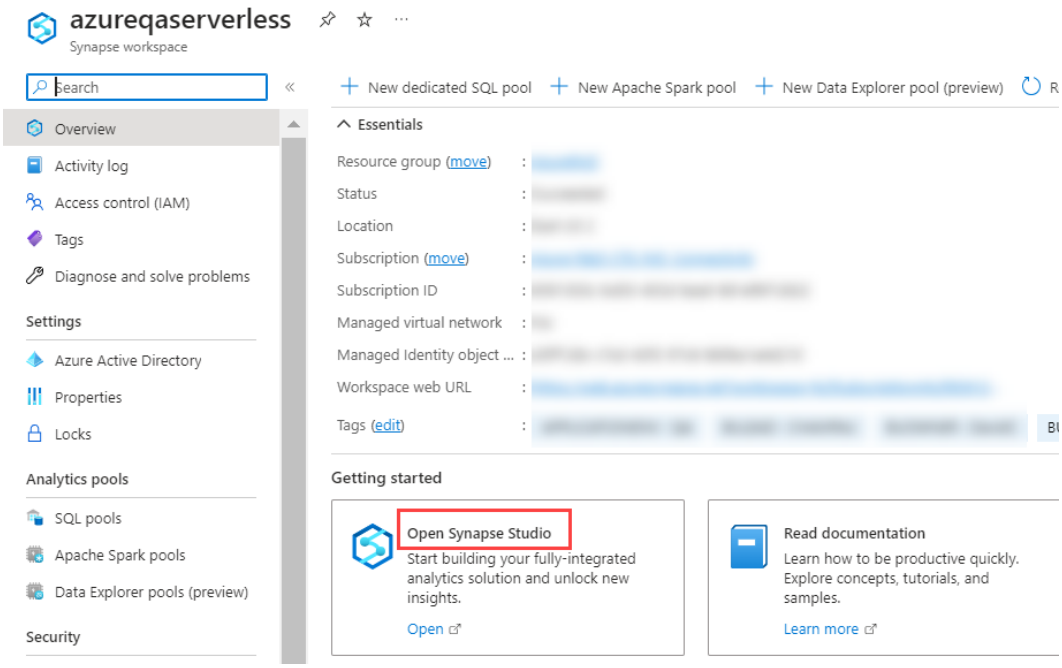


6. Select one of the following authentication methods:
   - **Use both local and Azure Active Directory (Azure AD) authentication**. Allows you to create a user for Azure AD or SQL authentication.
   - **Use only Azure Active Directory (Azure AD) authentication**. Allows you to create a user only for Azure AD authentication.
7. Specify the SQL admin credentials, and then click **Review + create**.

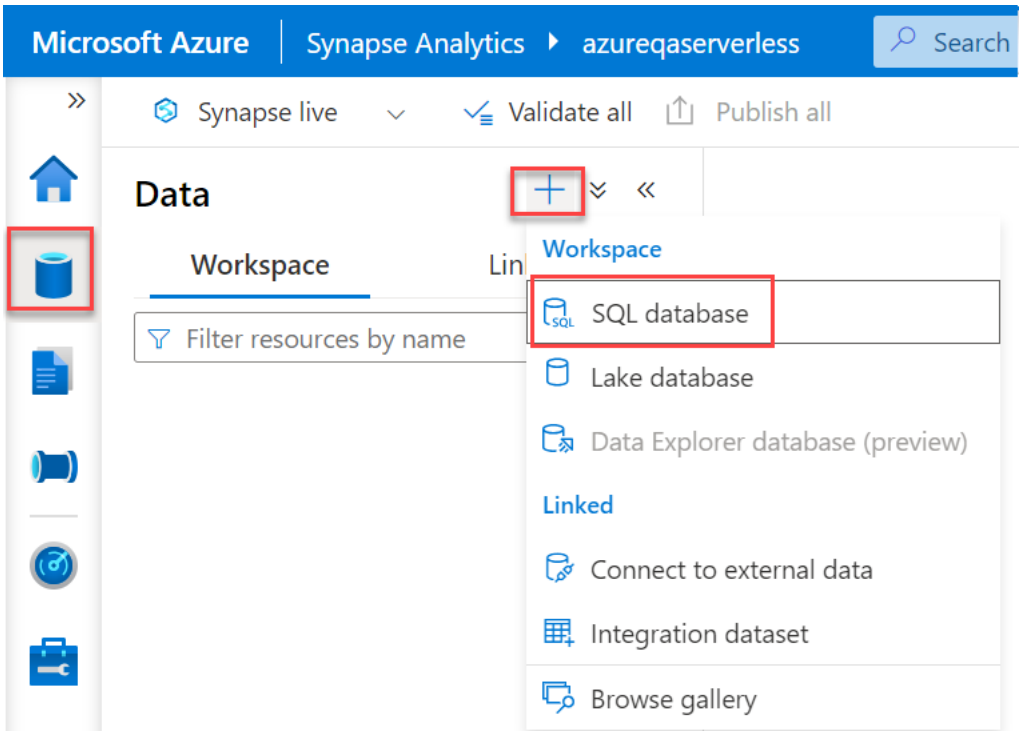
Creating an SQL database

Access your Azure account and create an SQL database.

1. On the Azure portal, navigate to the workspace you created.
2. Access **Open Synapse Studio**.

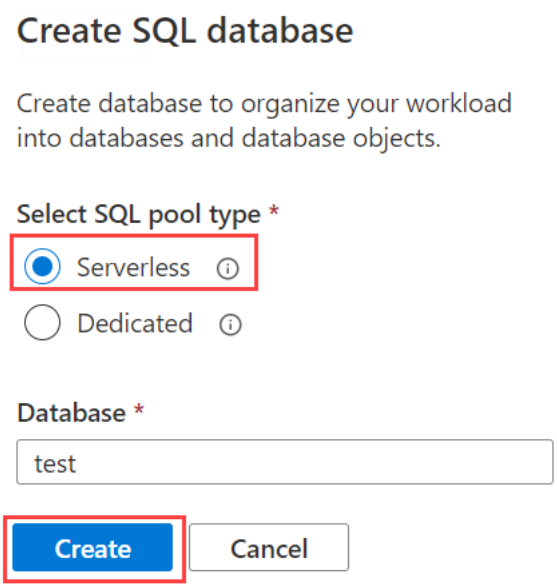


3. Navigate to **Data**, click **+**, and then click **SQL database**.



4. On the **Create SQL database** page, select **Serverless** as the SQL pool type.

5. Specify the database name, and then click **Create**.



Note: When you configure a Microsoft Azure Synapse SQL connection in Cloud Data Integration, ensure that the database points to the SQL database that you create and not to the master database.

Prepare for authentication

You can configure Azure Active Directory, Managed Identity, Microsoft SQL server, and Service Principal authentication types to connect to a serverless SQL pool.

Before you configure the authentication, be sure to complete the following prerequisites:

- Microsoft SQL server authentication. Requires you to create an SQL user.
- Azure Active Directory authentication. Requires you to grant privileges to an Azure AD user.
- Managed Identity authentication. Requires you to grant privileges for user assigned and system assigned identity.
- Service Principal authentication. Requires you to create a service principal user and assign the Storage Blob Data Contributor role.

The recommended authentication types to connect to a serverless SQL pool are Azure Active Directory and Managed Identity authentication.

Configuring Microsoft SQL server authentication

Configure Microsoft SQL Server authentication to connect to Microsoft Azure Synapse SQL.

1. Access Microsoft SQL Server Management Studio to connect to Microsoft Azure Synapse SQL using the Azure Active Directory admin credentials.

2. Create an SQL user. To do this, type and run the following command:

```
USE master
CREATE login <username> WITH PASSWORD = <password>
USE test2
CREATE USER <username> FOR LOGIN <username>
ALTER ROLE [db_owner] ADD MEMBER <username>
```

The SQL user must be a local user and must be assigned a db_owner role to execute queries.

Configuring Azure Active Directory (AAD) Authentication

Configure Azure Active Directory (AAD) Authentication to connect to Microsoft Azure Synapse SQL.

To configure the AAD authentication, perform the following tasks

Import the server certificate

If a trust store file is not configured for your organization and you want to use AAD authentication with Active Directory Federation Services in Azure, import the server certificate.

Import the server certificate to the following location:

```
<Secure Agent installation directory>\jdk\jre\lib\security\cacerts
```

Use the following command to import the certificate:

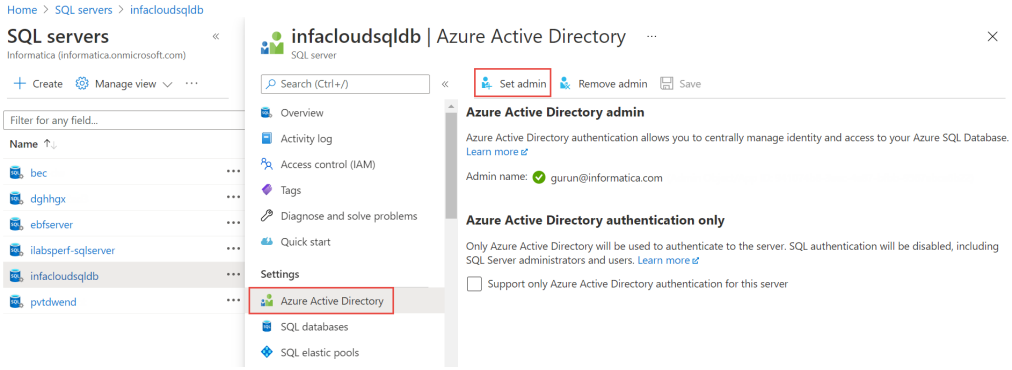
```
keytool -import -trustcacerts -alias <alias name of the certificate> -file <certificate file path> -keystore <Secure Agent installation directory>\jdk\jre\lib\security\cacerts -storepass <password for the truststore>
```

Set the Azure Active Directory admin

To add new users or delete existing users from your Azure Active Directory, you must have the administrator role.

Perform the following steps to set Azure Active Directory administrator:

1. Log in to the Azure portal.
2. On the All Resources page, select the Microsoft SQL Server that hosts Microsoft Azure Synapse SQL.
3. Select the **Azure Active Directory** option under Settings.
   The image shows the Azure Active Directory settings:



4. Click **Set admin**.
   The Add admin page appears.
5. Enter the email ID that you want to use as admin, and click **Select**.

6. Click **Save**.

Create a user

Perform the following steps to create a user:

1. Connect to Microsoft Azure Synapse SQL using the Azure Active Directory admin created in the previous steps.
   You can use Microsoft SQL Server Management Studio to connect to the Microsoft Azure Synapse SQL.
2. Run the following command:
   ```
   create user [user@foobar.com] from external provider;
   ```
   For more information, see the Microsoft Azure documentation.
3. To set an Azure Active Directory user as an Azure Active Directory admin and use the credentials, assign the following privileges to the user:
   ```
   CREATE USER [username] FROM EXTERNAL PROVIDER;
   ALTER ROLE db_datareader ADD MEMBER [username]
   ALTER ROLE db_datawriter ADD MEMBER [username]
   GRANT EXECUTE TO [username]
   grant ALTER ANY EXTERNAL DATA SOURCE to [username];
   grant create table to [username];
   grant create schema to [username];
   grant select to [username];
   grant update to [username];
   grant insert to [username];
   grant delete to [username];
   grant create view to [username];
   grant select on schema :: sys to [username];
   grant control to [username];
   EXEC sp_addrolemember 'db_owner','[username]';
   ALTER ROLE db_owner ADD MEMBER [username]
   ```

Configuring Managed Identity authentication

Before you use Managed Identity authentication, complete certain prerequisites.

1. Log in as an Azure Active Directory user.
2. Run the following command to grant permissions for System Assigned Identity:
   ```
   CREATE USER [MSITestLinux] FROM EXTERNAL PROVIDER; //Name of the Virtual Machine
   GRANT EXECUTE TO MSITestLinux
   grant ALTER ANY EXTERNAL DATA SOURCE to [MSITestLinux];
   grant create table to [MSITestLinux];
   grant create schema to [MSITestLinux];
   grant select to [MSITestLinux];
   grant update to [MSITestLinux];
   grant insert to [MSITestLinux];
   ```

```
grant delete to [MSITestLinux];

grant create view to [MSITestLinux];

grant select on schema :: sys to [MSITestLinux]

grant control to [MSITestLinux]
```

3. Select the Microsoft Azure Data Lake Storage Gen2 storage account and perform the following steps to assign the Storage Blob Contributor role to the system assigned identity:
   a. Click **Access control (IAM) > Add**.
   b. In the **Add role assignment** page, provide the **Storage Blob Contributor** role to the system assigned identity.
4. Run the following command to grant permissions for User Assigned Identity:

```
USE master

CREATE login [user] FROM EXTERNAL PROVIDER;

USE <database_name>

CREATE USER [user] for login [user]

ALTER ROLE db_datareader ADD MEMBER [user]

GRANT EXECUTE TO user

grant ALTER ANY EXTERNAL FILE FORMAT to [user];

grant ALTER ANY EXTERNAL DATA SOURCE to [user];

grant create table to [user];

grant create schema to [user];

grant select to [user];

grant update to [user];

grant insert to [user];

grant delete to [user];

grant create view to [user];

grant select on schema :: sys to [user];

grant control to [user]
```

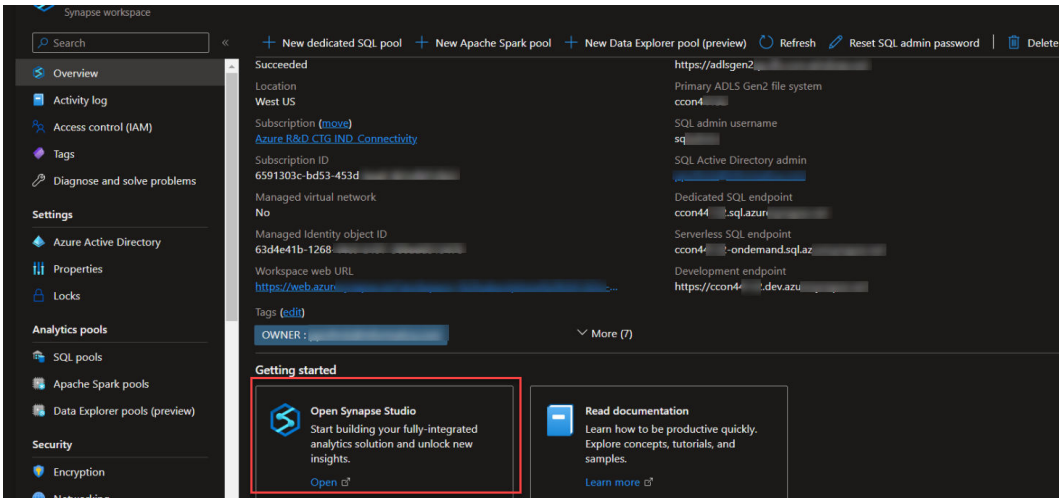
5. Select the Microsoft Azure Data Lake Storage Gen2 storage account and perform the following steps to assign the Storage Blob Contributor role to the user assigned identity:
   a. Click **Access control (IAM) > Add**.
   b. In the **Add role assignment** page, provide the **Storage Blob Contributor** role to the user assigned identity.
6. On the Azure portal, navigate to the workspace you created.
7. Access **Open Synapse Studio**.
8. Navigate to the Manage option on the left panel and click **Access control** to verify if the user is provided the access.

Configuring Service Principal authentication

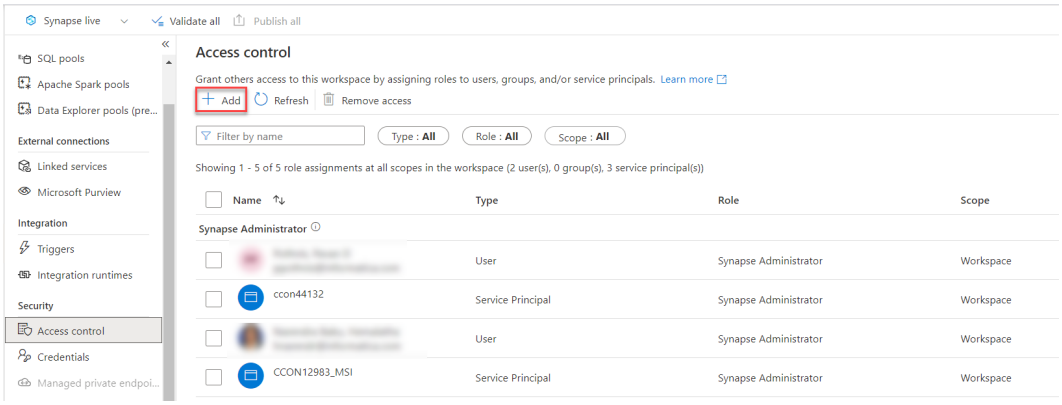
Create a service principal user for the serverless SQL pool and assign the Storage Blob Data Contributor role.

1. On the Azure portal, navigate to the serverless SQL pool.

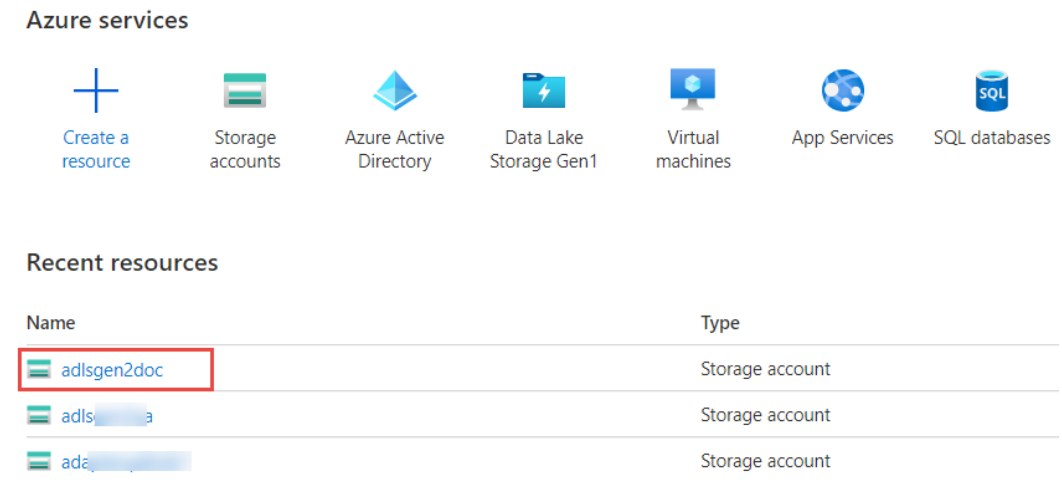
2. Access **Open Synapse Studio**.



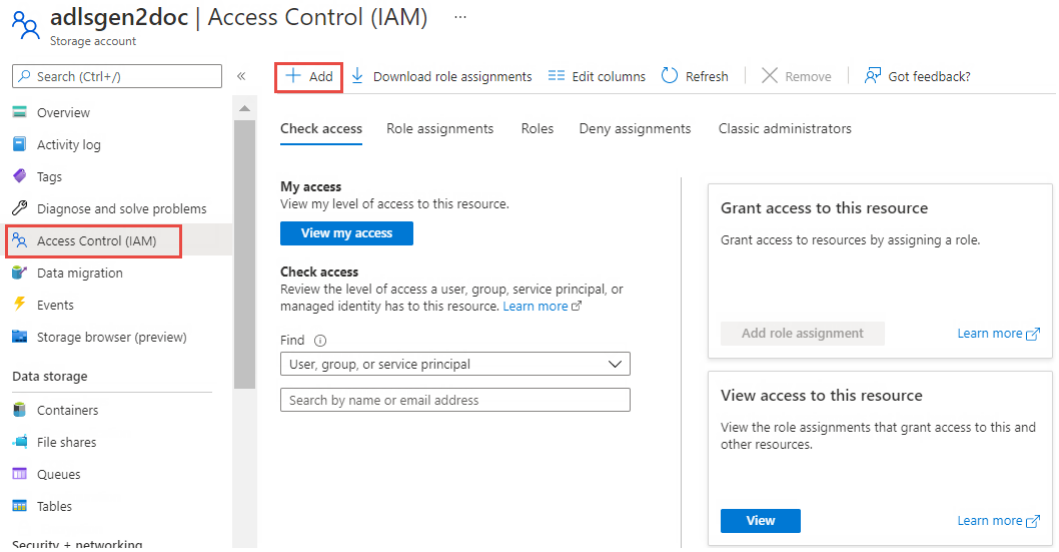
3. In the **Manage** section, click **Access control**.



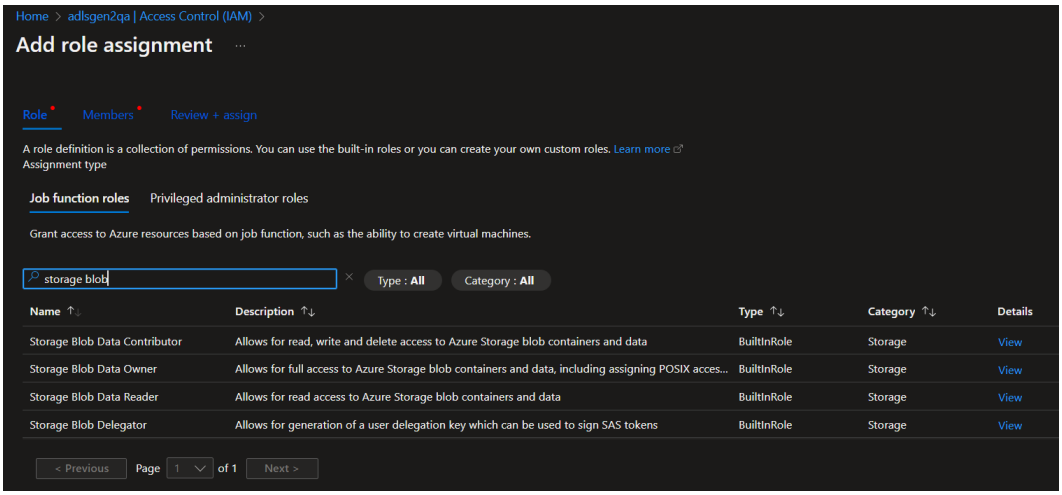
4. Click **Add** to add the service principal application.
5. Navigate to the Microsoft Azure Data Lake Storage Gen2 storage account.



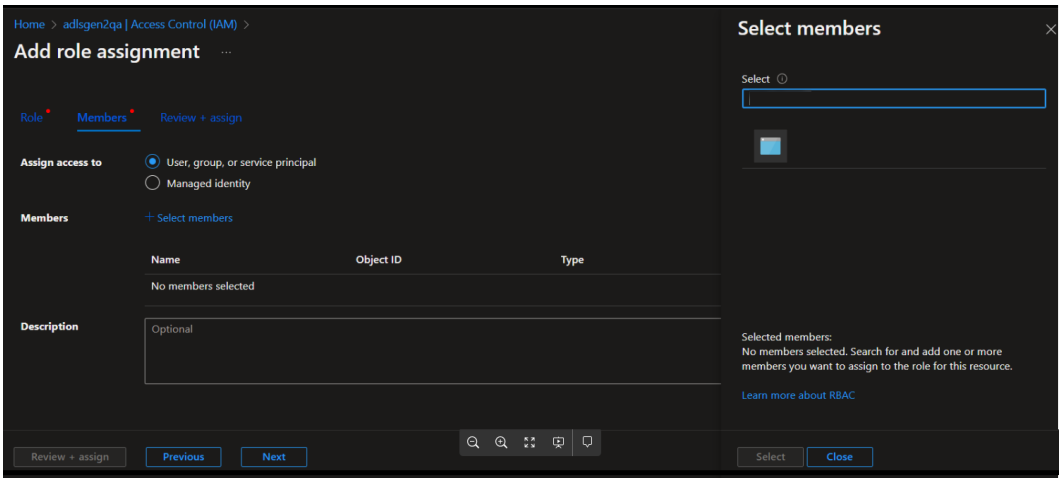
6. Click **Access control (IAM)** > **Add**.



7. In the **Add role assignment** page, select the **Role** tab, search for **Storage Blob Data Contributor** role, and then click **Next**.



8. On the **Members** tab, click **Select members** and search for the service principal application created earlier.



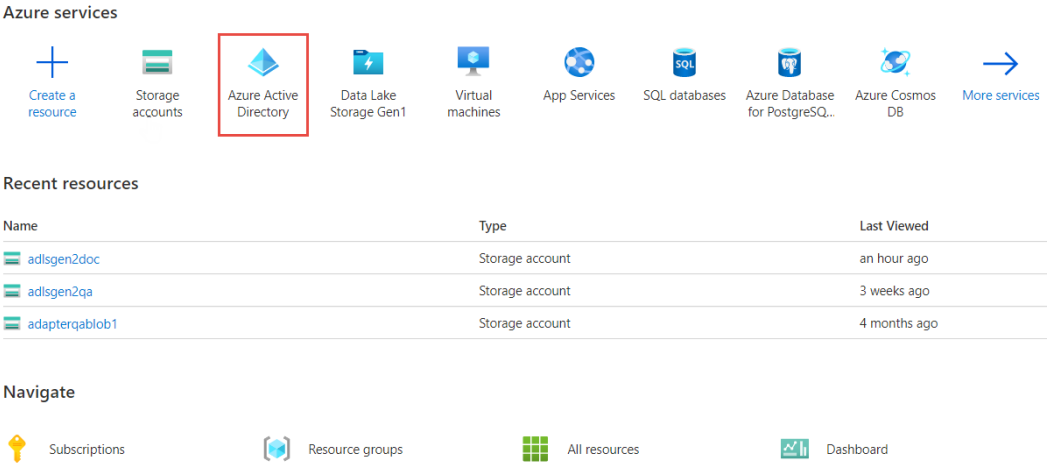
9. Select the service principal application and assign the role.

Obtaining credentials for Service Principal authentication

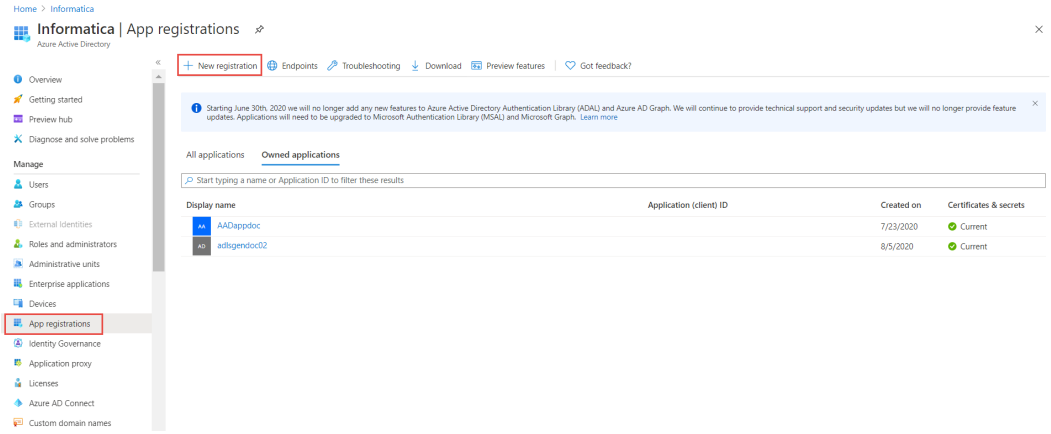
You can use service principal authentication to connect to Microsoft Azure Data Lake Storage Gen2 to stage files.

To use service principal authentication, register an application in the Azure Active Directory, generate a client secret, and then assign the Storage Blob Contributor role to the application.

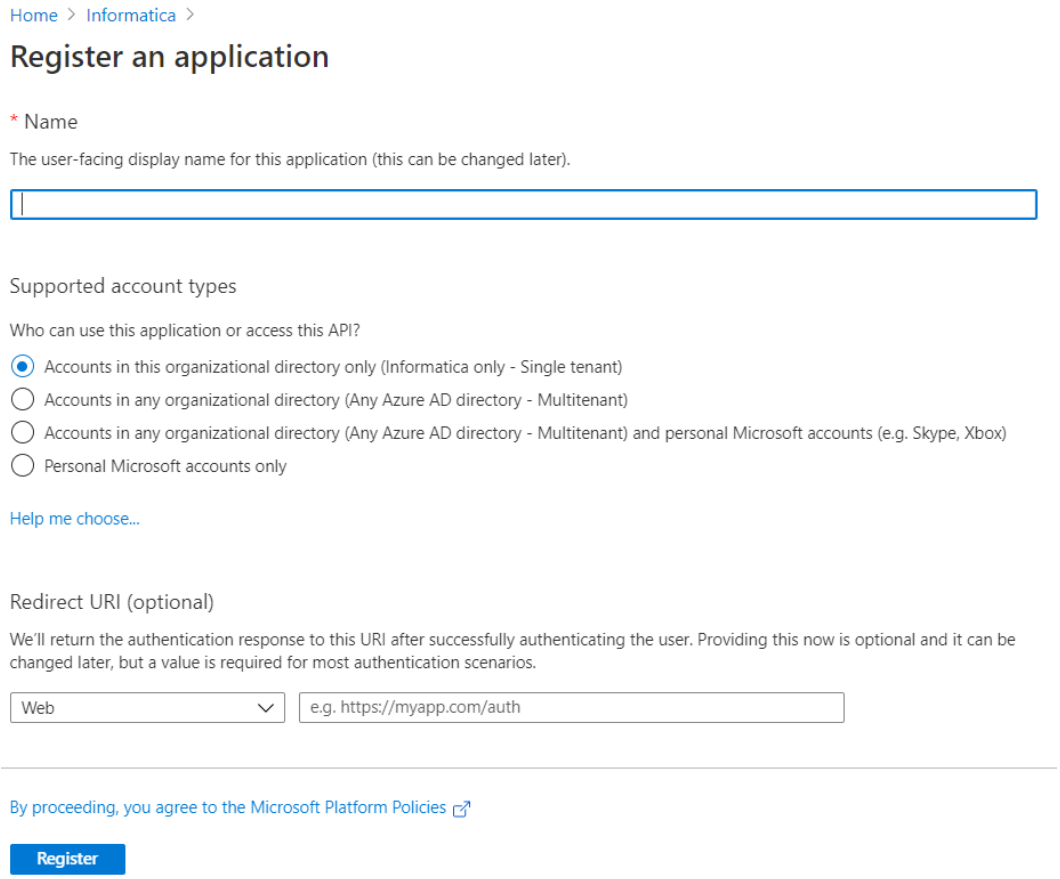
1. On the Azure portal, click **Azure Active Directory**.



2. In the **Manage** section, click **App registrations**.

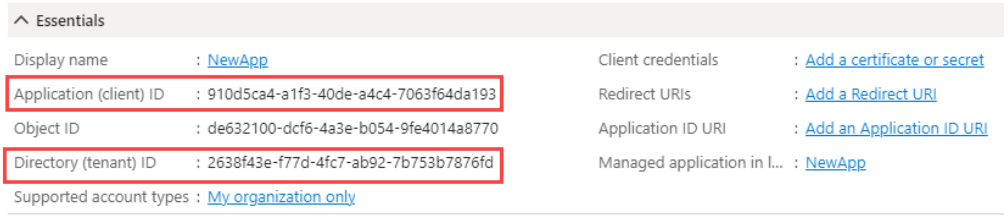


3. Click **New registration** to create a new Azure Active Directory application.
4. On the **Register an application** page, enter the details for the new application.

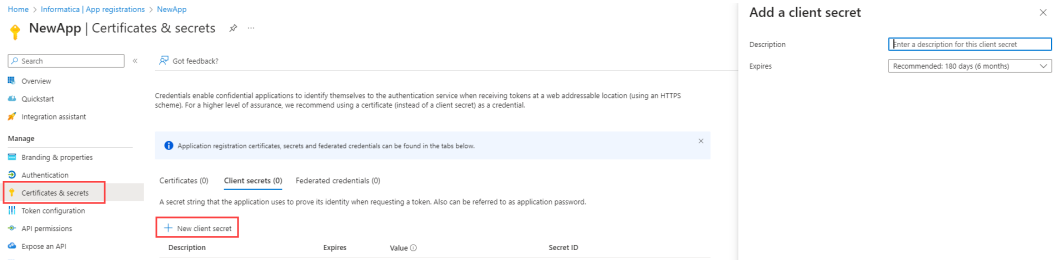


   a. In the **Name** field, enter the application name.
   b. In the **Redirect URI** section, select **Web** as the type of the application and enter the URL of the application.

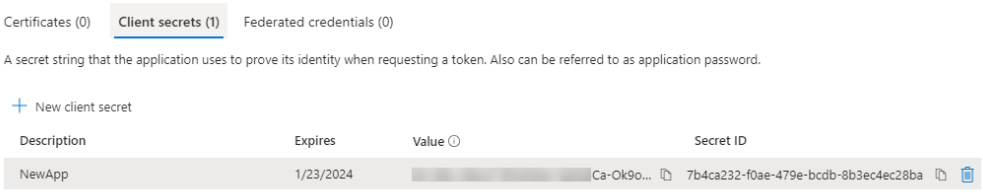
c. Click **Register**.
The details of the newly created Azure Active Directory application page are displayed.



5. In the Manage section, click **Certificates & secrets** section.
6. Click **New client secret**.

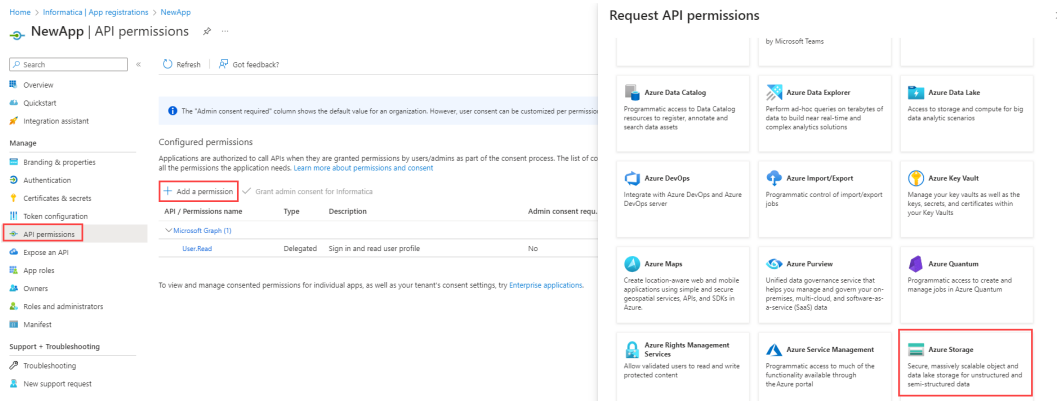


7. In the **Add a client secret** page, perform the following steps:
   a. Enter a name for the client secret in the **Description** field.
   b. In the **Expires** field, you can select the duration of the key as **Never**(Recommended).
   c. Click **Add**.
   d. The value of the key is generated and displayed in the **Value** field.

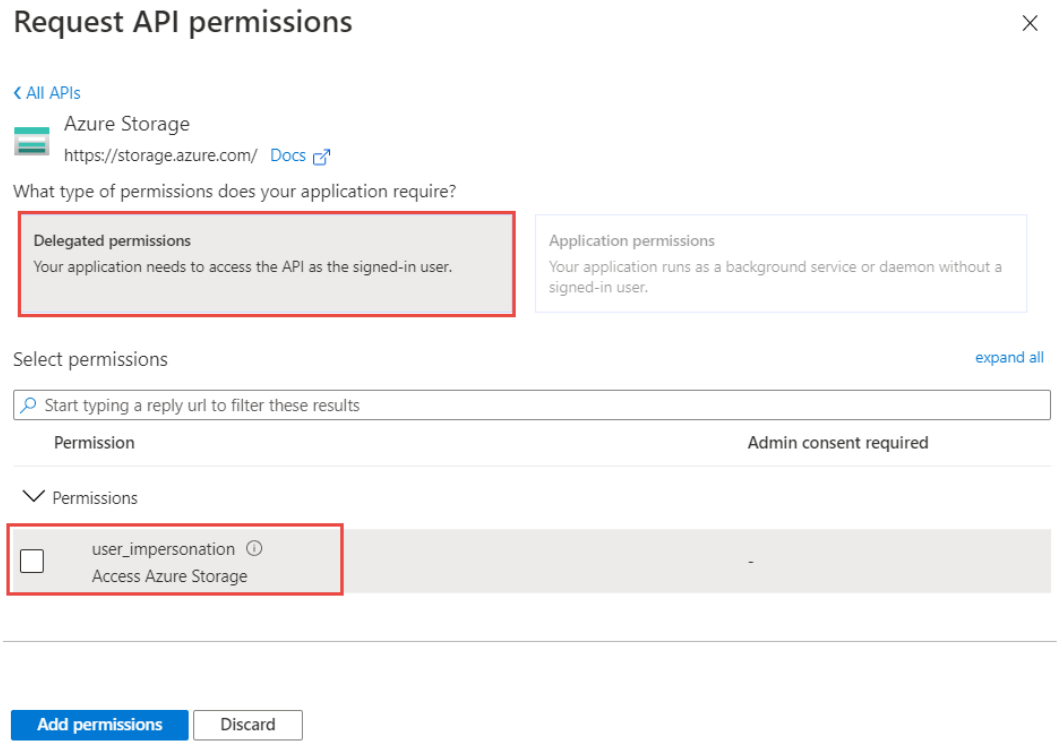


Note: You must copy the key value as you cannot retrieve the value after you leave the page. Ensure that the client secret does not contain special characters.

8. In the Manage section, click **Owners**.
9. Click **Add owner**.
10. In the **Search** field, search for the owner name or email address that you used to log in to the Azure portal.
11. Select the owner name or email address, and click **Select**.
12. In the Manage section, click **API permissions**.
The configured permissions are displayed.

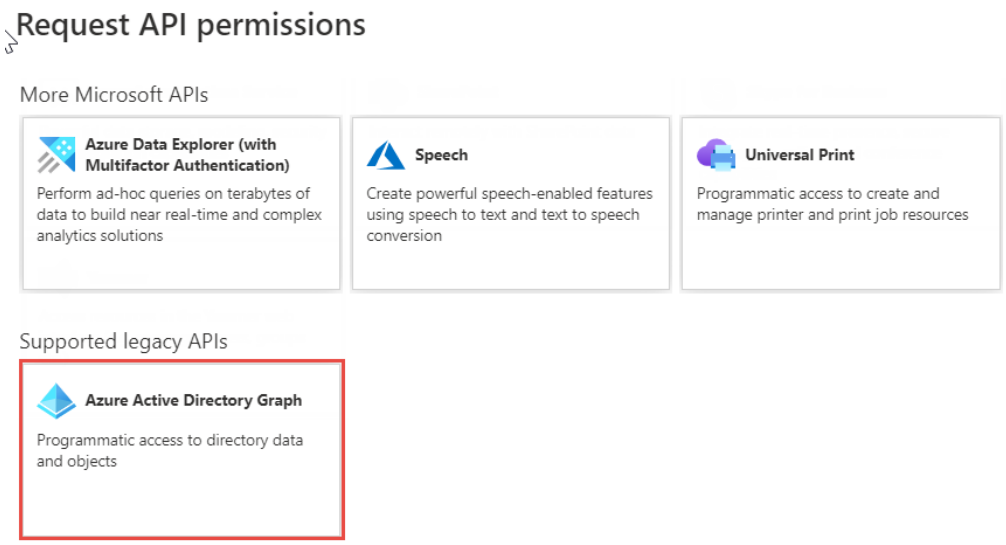


13. Click **Add a permission**.
    The **Request API permissions** page appears.
14. In the Microsoft APIs section, click **Azure Storage**.
15. Select **Delegated permissions** as the type of permissions.
16. Select **Access Azure Storage** from the listed permissions.



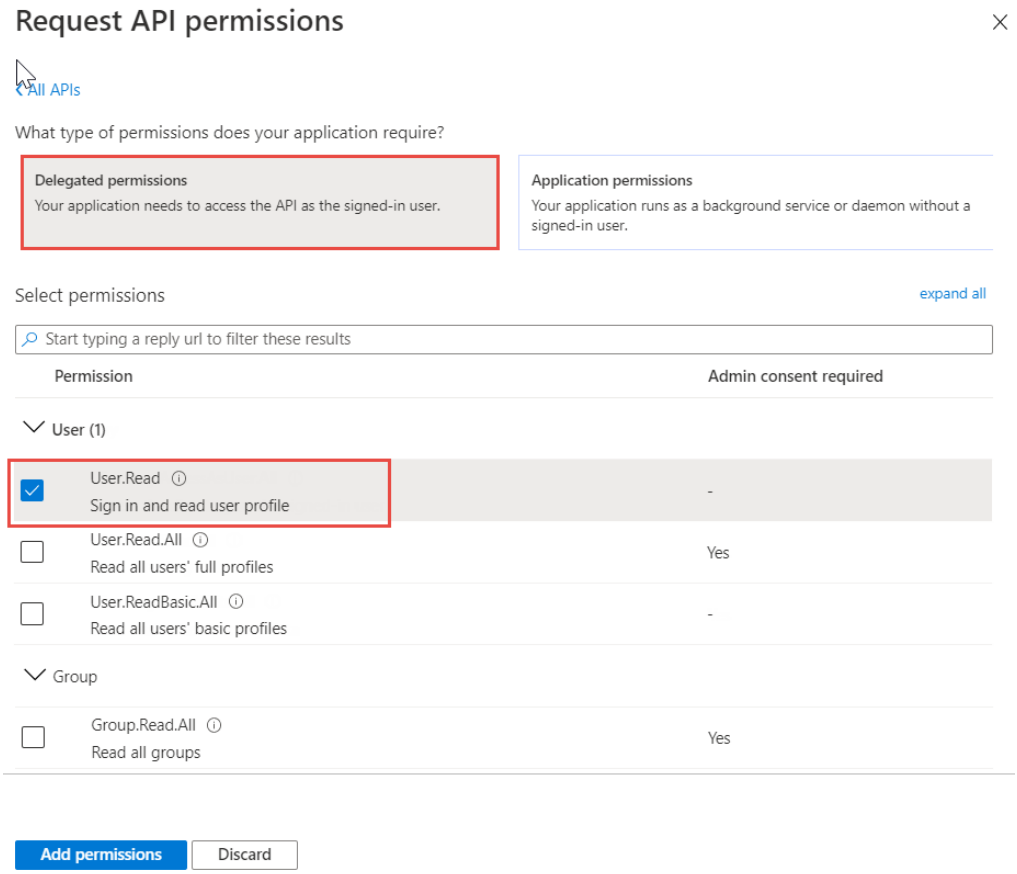
17. Click **Add permissions**.
18. In the **Configured permissions**, select **Azure Active directory** and ensure that the **Sign in and read user profile** option is enabled in the **Delegated permissions** section.
    If **Azure Active directory** is not listed under the **Configured permissions**, perform the following steps:
    a. Click **Add a permission**.
       The **Request API permissions** page appears.

b. In the Microsoft APIs section, click **Azure Active Directory Graph**.



c. Select **Delegated permissions** as the type of permissions.

d. Select **Sign in and read user profile** from the listed permissions.

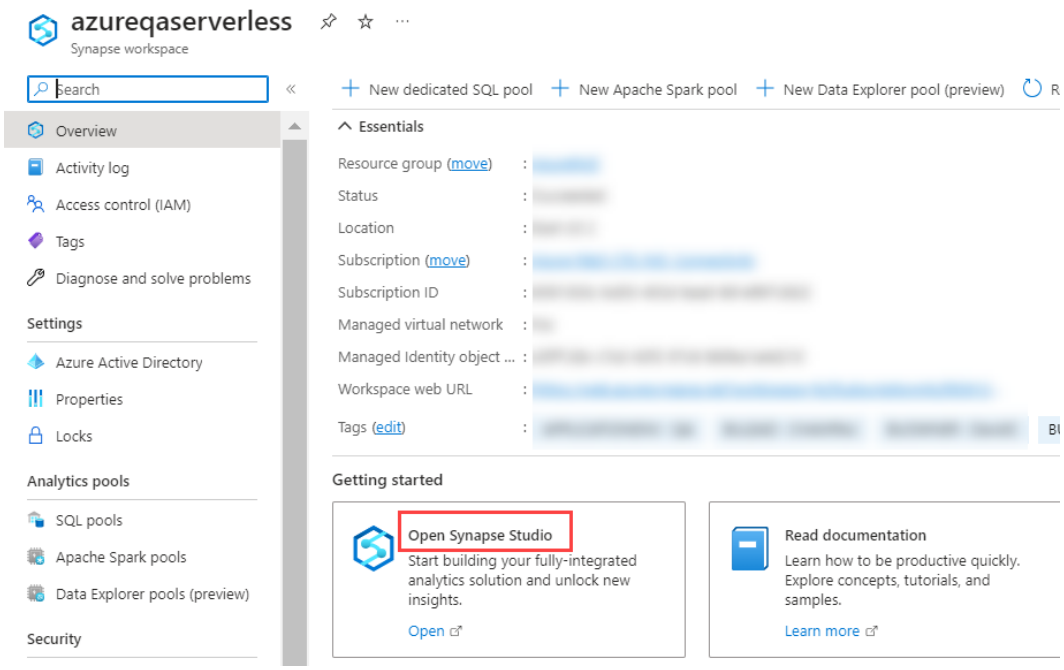


19. Go to the home page and in the Storage Account section, select the storage account that you created.
20. Click **Access control (IAM)** > **Add**.
21. In the **Add role assignment** page, provide the **Storage Blob Data Contributor** to the application.

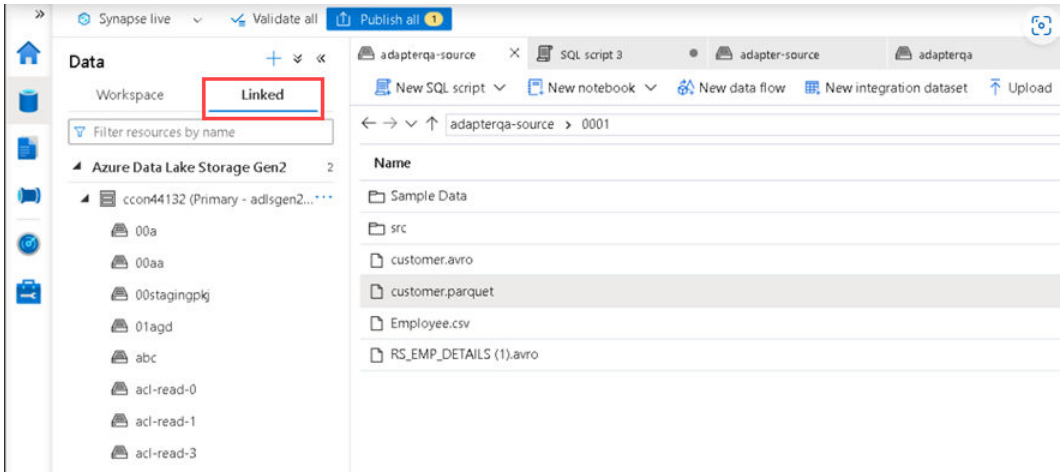
Reading data using OPENROWSET query

You can read data from a file in Microsoft Azure Data Lake Storage Gen2 using the OPENROWSET query.

1. On the Azure portal, navigate to the workspace you created.
2. Access **Open Synapse Studio**.



3. Navigate to **Data**, click the **Linked** tab.



4. Right-click a file and select **Select Top 100 rows** to use OPENROWSET query.

   If you configure Azure Active Directory or Managed Identity authentication, you can use the following command to create the query:

```
SELECT

TOP 100 *

FROM
```

```
OPENROWSET(
BULK '0001/customer.parquet',//Relative path
FORMAT = 'PARQUET'
) AS [result]
```

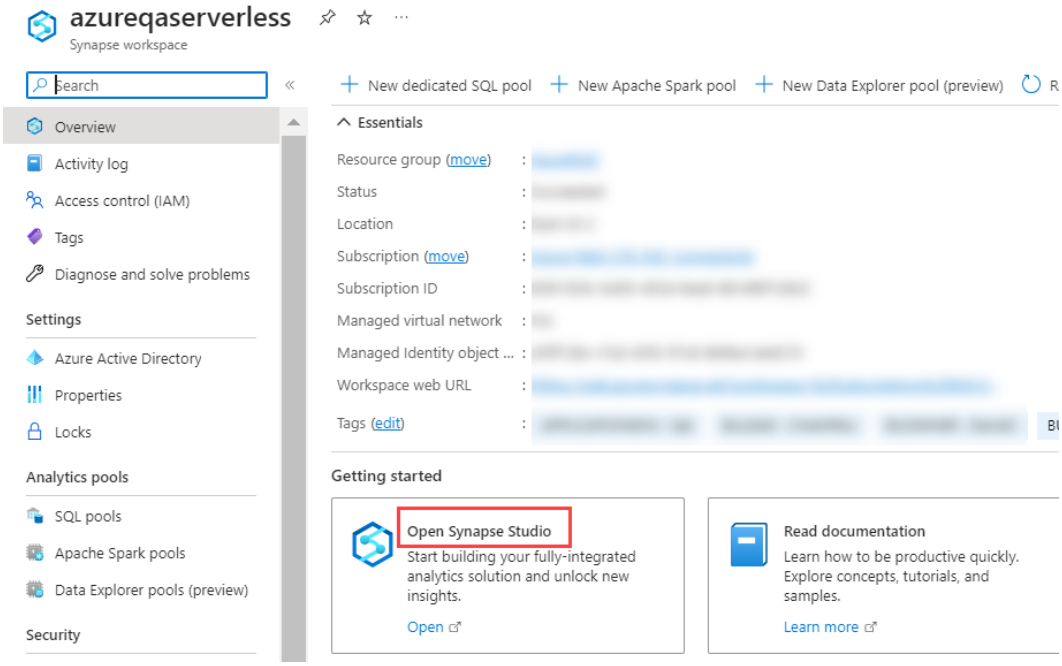
5. If you configure Microsoft SQL authentication, perform the following additional steps before you create the query:
    1. Access Microsoft SQL Server Management Studio, Azure Data Studio, or any other tool that allows access to Azure database and run the following command to create master key:
    ```
    IF NOT EXISTS (SELECT * FROM sys.symmetric_keys WHERE symmetric_key_id = 101)
    CREATE MASTER KEY
    ```
    2. Run the following command to create database scoped credentials:
    ```
    IF NOT EXISTS (SELECT * FROM sys.database_credentials WHERE NAME = '<Credential_Name>')
    BEGIN
    CREATE DATABASE SCOPED CREDENTIAL <Credential_Name>WITH IDENTITY = '<Identity_Name>',
    Secret = 'XXXXXX'
    END
    ```
    Only Service Principal authentication or Managed Identity authentication is applicable when you create database scoped credentials.
    3. Run the following command to create external data source:
    ```
    IF NOT EXISTS (SELECT * FROM sys.external_data_sources WHERE name = '<Data_Source_Name>')
    CREATE EXTERNAL DATA SOURCE <Data_Source_Name>
    WITH (
    LOCATION = '<Location>',
    CREDENTIAL = <Credential_Name>)//Credentials created in previous step
    ```
    4. Run the following command to create openrowset query:
    ```
    SELECT
    TOP 100 *
    FROM
    OPENROWSET(
    BULK '0001/customer.parquet',//Relative path
    DATA_SOURCE = 'adapterqa-source_adlsgen2qa_dfs_core_windows_net1',//Mandatory for SQL
    authentication.
    ```
    Ensure that the name of the data source matches with the name of the data source created in the previous step.
    ```
    FORMAT = 'PARQUET'
    ) AS [result]
    ```
    You cannot execute OPENROWSET queries on master database.

Reading data from a file using external table

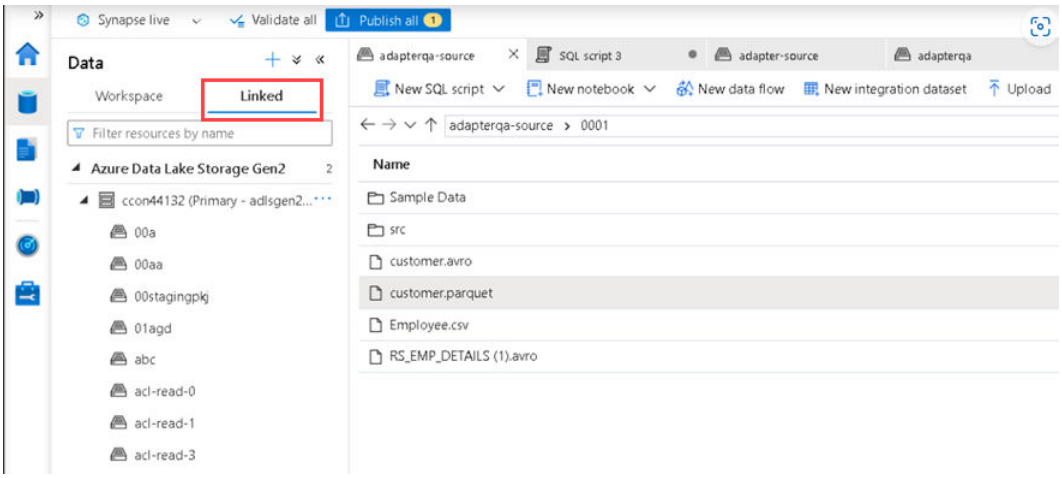
You can also read data from a file using an external table.

1. On the Azure portal, navigate to the workspace you created.

2. Access **Open Synapse Studio**.



3. Navigate to **Data**, click the **Linked** tab.



4. Right-click a file and select **Create external table**.
5. On the **New external table** page, select the following details:
   - Select SQL pool
   - Select a database
   - External table name
6. In the **Create external table** field, select **Automatically** or **Using SQL script**.

   If you configure Azure Active Directory or Managed Identity authentication, you can either run the script that is auto-generated or use the SQL script to create the external table.

7. If you configure Microsoft SQL authentication, you must use the SQL script and perform the following step to create database scoped credentials:

   - Run the following command in Microsoft SQL Server Management Studio, Azure Data Studio, or any other tool to create database scoped credentials with Service Principal authentication or Managed Identity authentication:

     ```
     IF NOT EXISTS (SELECT * FROM sys.database_credentials WHERE NAME = '<Credential_Name>')

     BEGIN

     CREATE DATABASE SCOPED CREDENTIAL <Credential_Name>WITH IDENTITY = '<Identity_Name>',

     Secret = 'XXXXXX'

     END
     ```

     Use the created database scoped credentials in the next steps to create the external table.

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